

Assignment: Loan Approval Prediction Using Machine Learning

Objective: Develop and evaluate machine learning models to predict whether a loan application should be approved based on applicant information. Through this assignment, apply Cross Validation, Hyperparameter Tuning, and Bias-Variance Tradeoff concepts.

Question 1. Load the Loan Prediction dataset and perform an initial analysis.

Tasks:

1. Display the first 10 records.
2. Identify all features and the target variable.
3. Determine the number of records and attributes.
4. Check for missing values.
5. Describe the statistical properties of numerical features.

Question 2. Perform data preprocessing.

Tasks:

1. Handle missing values using appropriate techniques.
2. Convert categorical variables into numerical format.
3. Normalize or standardize numerical features (if required).
4. Create the final feature matrix and target variable.

Question 3. Perform exploratory data analysis to understand factors affecting loan approval.

Create at least five visualizations, such as:

1. Loan approval distribution.
2. Applicant income vs loan approval.
3. Credit history vs loan approval.
4. Education level vs loan approval.
5. Property area vs loan approval.

Answer the following:

1. Which feature appears most influential in loan approval?
2. What patterns can be observed from the visualizations?

Question 4. Split the dataset into training and testing sets (80:20).

Train the following models:

1. Logistic Regression
2. Decision Tree Classifier
3. Random Forest Classifier

For each model: Calculate Accuracy, Precision, Recall, F1 Score. Present the results in a comparison table.

Question 5. Based on the performance metrics:

1. Identify the best-performing model.
2. Justify your choice using evaluation metrics.
3. Discuss the strengths and limitations of each model.

Question 6. Apply Stratified 5-Fold Cross Validation to all three models.

Tasks:

1. Compute accuracy for each fold.
2. Calculate mean accuracy and standard deviation.
3. Compare cross-validation results with the train-test split results.

Answer:

1. Why is Stratified K-Fold preferred for this dataset?
2. Which model shows the most consistent performance?

Question 7. Perform hyperparameter tuning on the Random Forest Classifier using GridSearchCV. Tune the following parameters:

`n_estimators = [50, 100, 200]`

`max_depth = [3, 5, 10]`

`min_samples_split = [2, 5, 10]`

Tasks:

1. Identify the best parameter combination.
2. Report the best cross-validation score.
3. Evaluate the tuned model on the test set.

Compare:

Model performance before tuning

Model performance after tuning

Question 8. Train three Decision Tree models using:

`max_depth = 2`

`max_depth = 5`

`max_depth = 15`

For each model:

1. Compute training accuracy.
2. Compute testing accuracy.
3. Compare the results.

Answer:

1. Which model exhibits underfitting?
2. Which model exhibits overfitting?
3. Which model provides the best balance between bias and variance?

Present your findings in a table.